...the cutoff frequencies ω_1, and ω_2 (in rad/sec) are

14. The voltage across the inductor (V_L) at resonance (in V) is

Prob. 4: *Consider the circuit shown in Figure 3*, and answer questions 15-17.

15. The current supplied by the source (I_1)

16. The voltage (**V0**)

17. The complex power supplied by the source is

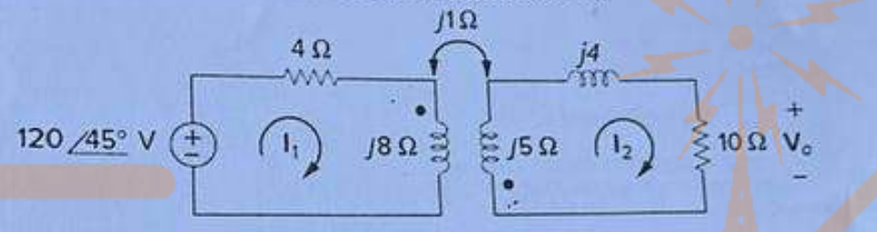


Figure 3

Prob. 5: Consider a balanced three phase Y- Y system has V_{an} = 440/ 0° V_{rms} in positive phase sequence and a load of Z_y = 6-j8 Ω. If the line impedance is zero as shown in figure 4 and answer questions 18-22.

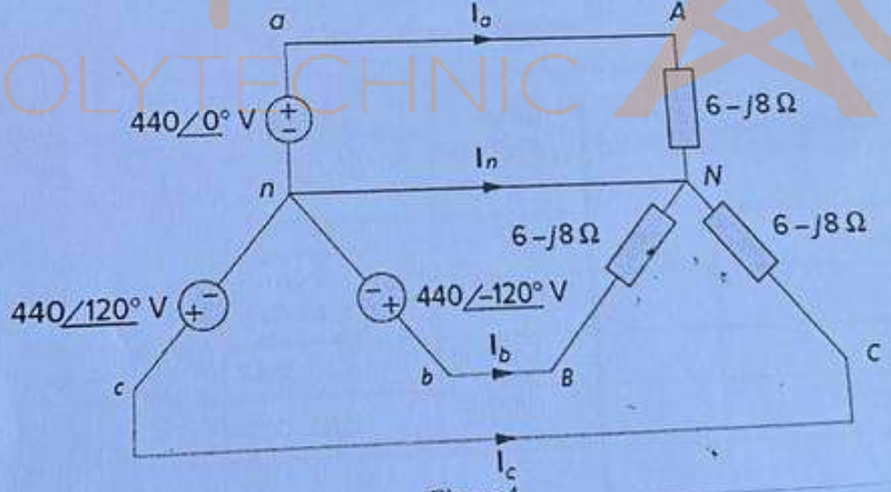


Figure 4

18. The voltage V_{AB} is:

19. The line current I_a is:

20. The phase current I_b is:

21. The total average power supplied by the source is:

22. The total complex power consumed by the load is:

23. The neutral line current is:

Good Luck

Write all the required equations (without simplification) to solve the following problems, then find the final answer.

Prob. 1: *Consider the circuit shown in Figure 1* **and answer questions 1-4.**

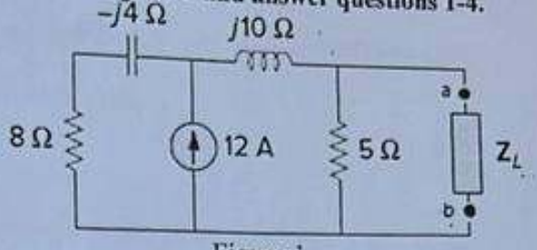


Figure 1

1. The Thevenin resistance with respect to terminals a-b is:
.....
2. The Thevenin voltage with respect to terminals a-b is:
.....
3. The load impedance **ZL** that absorbs the maximum average power is.
.....
4. The maximum average power absorbed by *ZL* is.
.....

Prob. 2: *Two Loads are connected in parallel across a voltage source of* $440\angle 0^\circ\ V_{rms}$ *as shown in figure 2.* **Answer questions 5-10.**

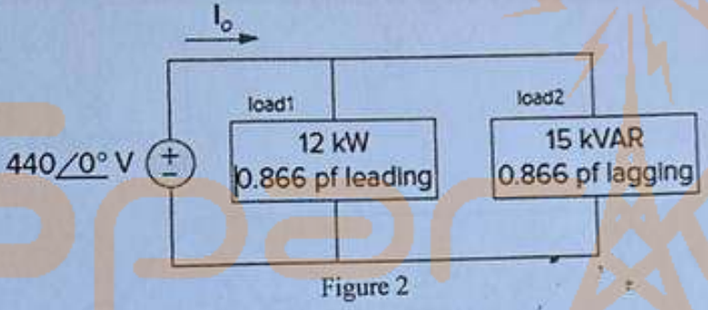


Figure 2

5. The current of load 1 is:
.....
6. The current of load 2 is:
.....
7. The complex power supplied by the source is:
.....
8. The power factor of the equivalent load is:
.....
9. The current supplied by the voltage source in rms is:
.....
10. The impedance of the equivalent load is:
.....

Prob. 3: *Consider a series RLC circuit has* $R = 5\ \Omega$, $L = 4mH$, $C = 0.1\mu F$, *and* $V_s = 10\angle 0^\circ\ V_{rms}$ **and answer questions 11-14.**

11. The resonant frequency (ω_o) in (rad/sec) is:
.....
12. The quality factor (**Q**) is:
.....